

Practical - 3

Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

A. Theory

1. How does the layout adapt for mobile, tablet, and desktop screens?

Answer: The layout changes using responsive CSS and media queries so it fits different screen sizes.

2. Which layout technique is used and why?

Answer: Flexbox and CSS Grid are used because they make layouts flexible, responsive, and easy to manage.

3. Are typography, spacing, colors, and contrast readable?

Answer: Yes. Proper font size, spacing, colors, and contrast make the content easy to read.

4. Is the framework used consistently without unnecessary inline CSS?

Answer: Yes. Bootstrap/Tailwind classes are used consistently and inline CSS is kept to a minimum.

B. CSS Code

```
*{  
  
margin:0;  
  
padding:0;  
  
box-sizing:border-box;  
  
font-family:Arial,Helvetica,sans-serif;  
}
```

```
body{  
  
background:#f5f7fb;  
  
color:#16233B;  
  
line-height:1.6;  
}
```

```
a{  
text-decoration:none;  
}
```

```
header{  
background:#16233B;  
padding:40px 20px;  
text-align:center;  
position:relative;  
overflow:hidden;  
}
```

```
header::before{  
content:"";  
position:absolute;  
width:250px;  
height:250px;  
background:rgba(255,255,255,.05);  
border-radius:50%;  
left:-80px;  
top:-80px;  
}
```

```
header::after{
```

```
content:"";  
position:absolute;  
width:220px;  
height:220px;  
background:rgba(255,255,255,.05);  
border-radius:50%;  
right:-60px;  
bottom:-80px;  
}
```

```
header h1{  
color:#fff;  
font-size:42px;  
margin-bottom:15px;  
}
```

```
header h2{  
color:#fff;  
font-size:28px;  
font-weight:400;  
margin-top:25px;  
}
```

```
nav{  
margin-top:20px;
```

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```
text-align:center;
```

```
}
```

```
nav a{
```

```
display:inline-block;
```

```
color:#fff;
```

```
padding:12px 22px;
```

```
margin:5px;
```

```
border-radius:30px;
```

```
font-weight:bold;
```

```
transition:.3s;
```

```
}
```

```
nav a:hover{
```

```
background:#C89B3C;
```

```
color:#16233B;
```

```
}
```

```
body>h1,
```

```
body>h2{
```

```
text-align:center;
```

```
margin:35px 0;
```

```
color:#16233B;
```

```
}
```

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```
body>nav{  
background:#16233B;  
padding:18px;  
margin-bottom:35px;  
}
```

```
body>nav a{  
color:#fff;  
}
```

```
body>nav a:hover{  
background:#C89B3C;  
color:#16233B;  
}
```

```
form{  
width:420px;  
max-width:92%;  
margin:30px auto;  
background:#fff;  
padding:35px;  
border-radius:18px;  
box-shadow:0 10px 30px rgba(0,0,0,.08);  
}
```

```
label{  
  display:block;  
  margin-bottom:8px;  
  font-size:17px;  
  font-weight:bold;  
  color:#16233B;  
}
```

```
input{  
  width:100%;  
  padding:14px;  
  border:1px solid #d8dce5;  
  border-radius:10px;  
  margin-bottom:20px;  
  font-size:15px;  
  transition:.3s;  
}
```

```
input:focus{  
  outline:none;  
  border-color:#C89B3C;  
  box-shadow:0 0 8px rgba(200,155,60,.3);  
}
```

```
button{
```

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```
padding:12px 28px;  
border:none;  
background:#C89B3C;  
color:#16233B;  
font-size:16px;  
font-weight:bold;  
border-radius:30px;  
cursor:pointer;  
transition:.3s;  
}
```

```
button:hover{  
background:#16233B;  
color:#fff;  
}
```

```
form+p{  
text-align:center;  
margin-top:20px;  
font-size:16px;  
color:#16233B;  
}
```

```
form+p+a{  
display:block;
```

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```
text-align:center;  
margin-top:10px;  
font-weight:bold;  
color:#16233B;  
}
```

```
form+p+a:hover{  
color:#C89B3C;  
}
```

```
p{  
margin-bottom:15px;  
padding:0 20px;  
}
```

```
ul{  
margin:20px 50px;  
}
```

```
li{  
margin-bottom:10px;  
}
```

```
table{  
width:95%;
```

```
margin:30px auto;  
border-collapse:collapse;  
background:#fff;  
box-shadow:0 10px 25px rgba(0,0,0,.08);  
}
```

```
th{  
background:#16233B;  
color:#fff;  
padding:14px;  
}
```

```
td{  
padding:12px;  
text-align:center;  
border:1px solid #ddd;  
}
```

```
tr:nth-child(even){  
background:#f9f9f9;  
}
```

```
tr:hover{  
background:#fff7e6;  
}
```

```
textarea{  
width:100%;  
padding:12px;  
border:1px solid #ddd;  
border-radius:10px;  
resize:vertical;  
}
```

```
textarea:focus{  
outline:none;  
border-color:#C89B3C;  
}
```

```
pre{  
background:#fff;  
padding:15px;  
margin:12px 20px;  
border-left:5px solid #C89B3C;  
border-radius:10px;  
box-shadow:0 5px 15px rgba(0,0,0,.08);  
white-space:pre-wrap;  
}
```

```
div{
```

```
background:#fff;
padding:20px;
margin:20px;
border-radius:15px;
box-shadow:0 8px 20px rgba(0,0,0,.08);
}
```

```
div a{
display:inline-block;
margin-top:12px;
background:#16233B;
color:#fff;
padding:10px 18px;
border-radius:8px;
}
```

```
div a:hover{
background:#C89B3C;
color:#16233B;
}
```

```
section{
background:#fff;
padding:25px;
margin:20px;
```

```
border-radius:15px;  
box-shadow:0 8px 20px rgba(0,0,0,.08);  
}
```

```
section a{  
display:inline-block;  
margin-top:15px;  
background:#16233B;  
color:#fff;  
padding:10px 18px;  
border-radius:8px;  
}
```

```
section a:hover{  
background:#C89B3C;  
color:#16233B;  
}
```

```
img{  
max-width:100%;  
display:block;  
}
```

```
footer{  
background:#16233B;
```

```
color:#fff;  
padding:20px;  
text-align:center;  
margin-top:40px;  
}
```

```
@media(max-width:768px){  
header h1{  
font-size:34px;  
}
```

```
header h2{  
font-size:22px;  
}
```

```
nav a{  
display:block;  
width:200px;  
margin:8px auto;  
}
```

```
table{  
font-size:14px;  
}
```

```
form{  
padding:25px;  
}  
  
main{  
width:95%;  
margin:30px auto;  
display:grid;  
grid-template-columns:repeat(auto-fit,minmax(260px,1fr));  
gap:25px;  
}
```

```
section{  
transition:.3s;  
cursor:pointer;  
}
```

```
section:hover{  
transform:translateY(-6px);  
box-shadow:0 15px 35px rgba(0,0,0,.15);  
}
```

```
section h3{  
margin-bottom:10px;  
font-size:22px;
```

```
color:#16233B;
```

```
}
```

```
section p{
```

```
padding:0;
```

```
margin-bottom:18px;
```

```
color:#666;
```

```
}
```

```
section a{
```

```
font-weight:bold;
```

```
}
```

```
div{
```

```
transition:.3s;
```

```
}
```

```
div:hover{
```

```
transform:translateY(-6px);
```

```
box-shadow:0 15px 35px rgba(0,0,0,.15);
```

```
}
```

```
div h3{
```

```
margin-bottom:10px;
```

```
color:#16233B;
```

```
}
```

```
div p{  
padding:0;  
color:#666;  
}
```

```
pre{  
font-size:16px;  
line-height:2;  
}
```

```
textarea:hover,  
input:hover{  
border-color:#C89B3C;  
}
```

```
table{  
overflow:hidden;  
border-radius:12px;  
}
```

```
th{  
letter-spacing:1px;  
}
```

```
td{  
transition:.2s;  
}
```

```
tr:hover td{  
color:#16233B;  
font-weight:bold;  
}
```

```
button{  
box-shadow:0 5px 15px rgba(0,0,0,.15);  
}
```

```
button:active{  
transform:scale(.98);  
}
```

```
nav a{  
position:relative;  
}
```

```
nav a::after{  
content:"";  
position:absolute;
```

```
left:15%;  
bottom:6px;  
width:0;  
height:2px;  
background:#fff;  
transition:.3s;  
}
```

```
nav a:hover::after{  
width:70%;  
}
```

```
body{  
animation:fade .7s;  
}
```

```
@keyframes fade{  
from{  
opacity:0;  
transform:translateY(20px);  
}  
to{  
opacity:1;  
transform:translateY(0);  
}
```

```
}
```

```
@media(max-width:900px){
```

```
main{
```

```
grid-template-columns:1fr;
```

```
}
```

```
section,
```

```
div{
```

```
margin:10px;
```

```
}
```

```
pre{
```

```
font-size:14px;
```

```
}
```

```
}
```

```
@media(max-width:600px){
```

```
body>h1{
```

```
font-size:28px;
```

```
}
```

```
body>h2{  
font-size:22px;  
}
```

```
table{  
display:block;  
overflow-x:auto;  
}
```

```
button{  
width:100%;  
}
```

```
textarea{  
width:100%;  
}
```

```
}  
h1,h2,h3{  
letter-spacing:.5px;  
}
```

```
h1{  
text-shadow:1px 1px 2px rgba(0,0,0,.1);  
}
```

```
form{  
transition:.3s;  
}
```

```
form:hover{  
box-shadow:0 15px 35px rgba(0,0,0,.15);  
transform:translateY(-3px);  
}
```

```
input,  
textarea,  
button{  
font-family:Arial,Helvetica,sans-serif;  
}
```

```
input,  
textarea{  
background:#fff;  
}
```

```
input::placeholder,  
textarea::placeholder{  
color:#999;  
}
```

```
input:focus,  
textarea:focus{  
background:#ffff7;  
}
```

```
label{  
margin-top:10px;  
}
```

```
ul{  
background:#fff;  
padding:25px 45px;  
border-radius:15px;  
box-shadow:0 8px 20px rgba(0,0,0,.08);  
}
```

```
li::marker{  
color:#C89B3C;  
}
```

```
table{  
transition:.3s;  
}
```

```
table:hover{  
  box-shadow:0 15px 35px rgba(0,0,0,.15);  
}
```

```
th{  
  font-size:16px;  
}
```

```
td{  
  font-size:15px;  
}
```

```
pre strong{  
  color:#16233B;  
}
```

```
section,  
div{  
  border-top:4px solid #C89B3C;  
}
```

```
section:hover,  
div:hover{  
  border-top-color:#16233B;  
}
```

```
section a,  
div a{  
transition:.3s;  
}
```

```
section a:hover,  
div a:hover{  
transform:translateY(-2px);  
}
```

```
button{  
transition:.3s;  
}
```

```
button:hover{  
box-shadow:0 8px 18px rgba(0,0,0,.2);  
}
```

```
button:active{  
transform:scale(.97);  
}
```

```
nav{  
position:sticky;
```

```
top:0;  
z-index:100;  
}
```

```
body>nav{  
box-shadow:0 4px 10px rgba(0,0,0,.1);  
}
```

```
hr{  
border:none;  
height:1px;  
background:#ddd;  
margin:25px 0;  
}
```

```
body{  
min-height:100vh;  
}
```

```
@media(max-width:768px){
```

```
ul{  
margin:20px;  
padding:20px;  
}
```

```
table th,  
table td{  
padding:10px;  
}
```

```
pre{  
margin:10px;  
}
```

```
}
```

```
@media(max-width:500px){
```

```
header{  
padding:30px 15px;  
}
```

```
header h1{  
font-size:28px;  
}
```

```
header h2{  
font-size:20px;  
}
```

```
body>h1{  
font-size:26px;  
}
```

```
body>h2{  
font-size:20px;  
}
```

```
nav a{  
padding:10px 16px;  
font-size:14px;  
}
```

```
form{  
width:95%;  
padding:20px;  
}
```

```
table{  
font-size:13px;  
}
```

```
button{  
padding:10px 20px;
```

```
}
```

```
}
```

```
.logo-title{  
    display:flex;  
    justify-content:center;  
    align-items:center;  
    gap:15px;  
    margin-bottom:20px;  
    position:relative;  
    z-index:2;  
}
```

```
.logo{  
    width:200px;  
    height:200px;  
    object-fit:contain;  
    margin:0;  
}
```

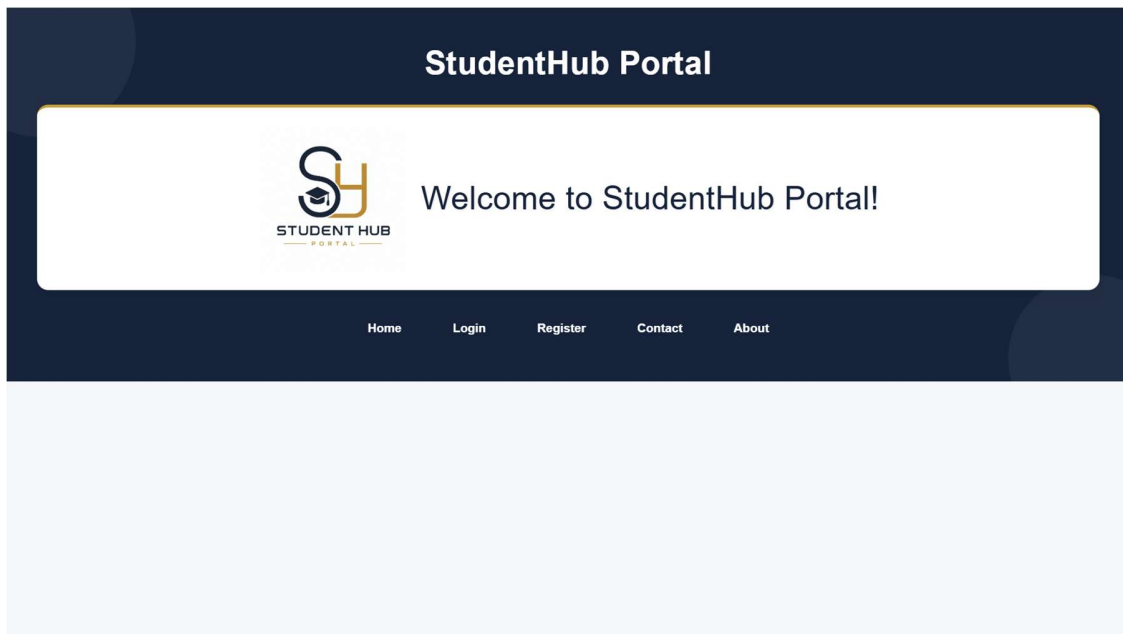
```
.logo-title h1{  
    margin:0;  
    color: #16233B;;  
    font-size:42px;  
}
```

```
.home-title{
```

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```
color:#16233B;  
font-size:28px;  
font-weight:400;  
margin-top:25px;  
}
```

D. Evidence of Work (Screenshots)



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Welcome Back!

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Welcome!

Home

Email ID:

Mobile Number:

Password:

Confirm Password:

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Email ID:

Mobile Number:

Password:

Confirm Password:

Sign Up

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Time Table

View your Time Table.

View Time Table

Study Material & Courses

Access your course study materials.

View Study Material

Assignments

View and manage your assignments.

View Assignments

Results

Check your academic results.

View Results

Department of Computer Engineering

Time Table

[Dashboard](#) [Time Table](#) [Assignment](#) [Courses](#) [Result](#)

Time	Monday	Tuesday	Wednesday	Thursday	Friday
9:00 - 10:00	WDF	DSA	OOP	Maths	CN
10:00 - 11:00	OOP	WDF	CN	DSA	Maths
11:00 - 12:00			Break		
12:00 - 01:00	Maths	CN	Maths	OOP	DSA
11:00 - 12:00	DSA	OOP	WDF	CN	WDF

Assignment

[Dashboard](#) [Time Table](#) [Assignment](#) [Courses](#) [Result](#)

Subject	Assignment	Due Date	Status
Web Development	HTML Website	10 August 2026	Submitted
Data Structures and Algorithm	Binary Search	11 August 2026	Submitted
Object Oriented Programming	Inheritance	11 August 2026	Done Late
Computer Networking	Tologies	12 August 2026	Submitted
Discrete Mathematics	Unit 1	14 August 2026	Pending

Department of Computer Engineering

Courses

[Dashboard](#)[Time Table](#)[Assignment](#)[Courses](#)[Result](#)

Study Material

Course 1

Course Name: Web Development

[View Material](#)

Course 2

Course Name: Data Structures and Algorithm

[View Material](#)

Course 3

Course Name: Object Oriented Programming

[Dashboard](#)[Time Table](#)[Assignment](#)[Courses](#)[Result](#)

Course Name: Data Structures and Algorithm

[View Material](#)

Course 3

Course Name: Object Oriented Programming

[View Material](#)

Course 4

Course Name: Computer Networks

[View Material](#)

Course 5

Course Name: Discrete Mathematics

[View Material](#)

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Result

[Dashboard](#)[Time Table](#)[Assignment](#)[Courses](#)[Result](#)

Academic Results

Student Name: Student Name

Student ID: STU001

Subject	Internal Marks	External Marks	Total	Grade
Web Development	24	56	80	A
Data Structures and Algorithm	24	64	88	A+
Object Oriented Programming	29	68	97	O
Computer Networks	26	66	92	A+
Discrete Mathematics	24	60	84	A

Download Result

Profile

[Dashboard](#)[Profile](#)

Student Information

Name

: Student Name

Student ID

: STU001

Email

: student@example.com

Mobile

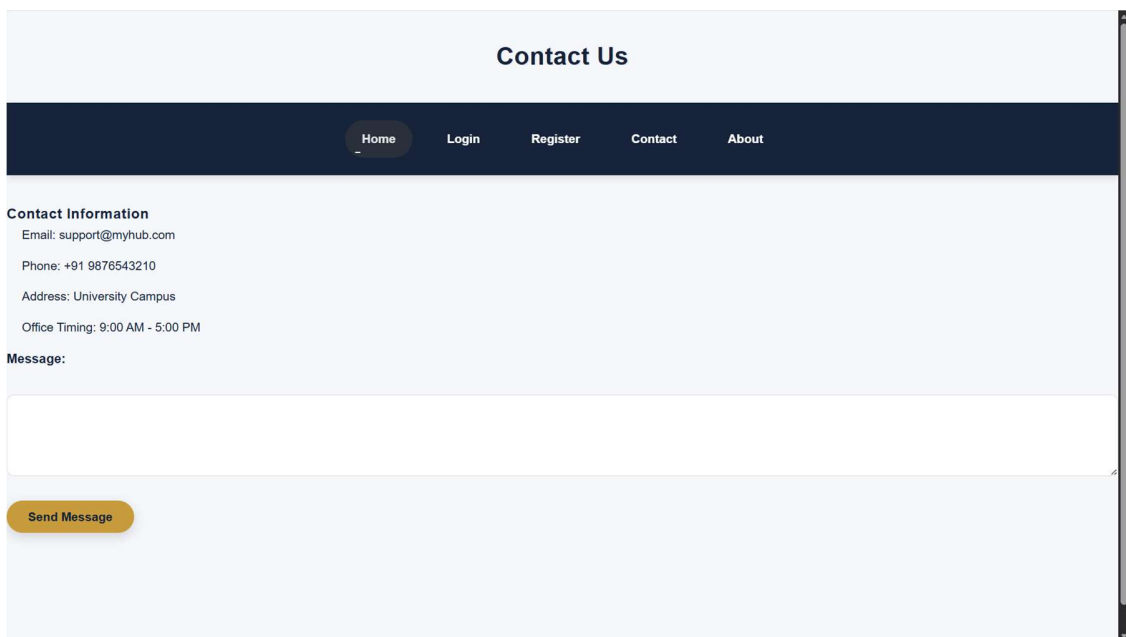
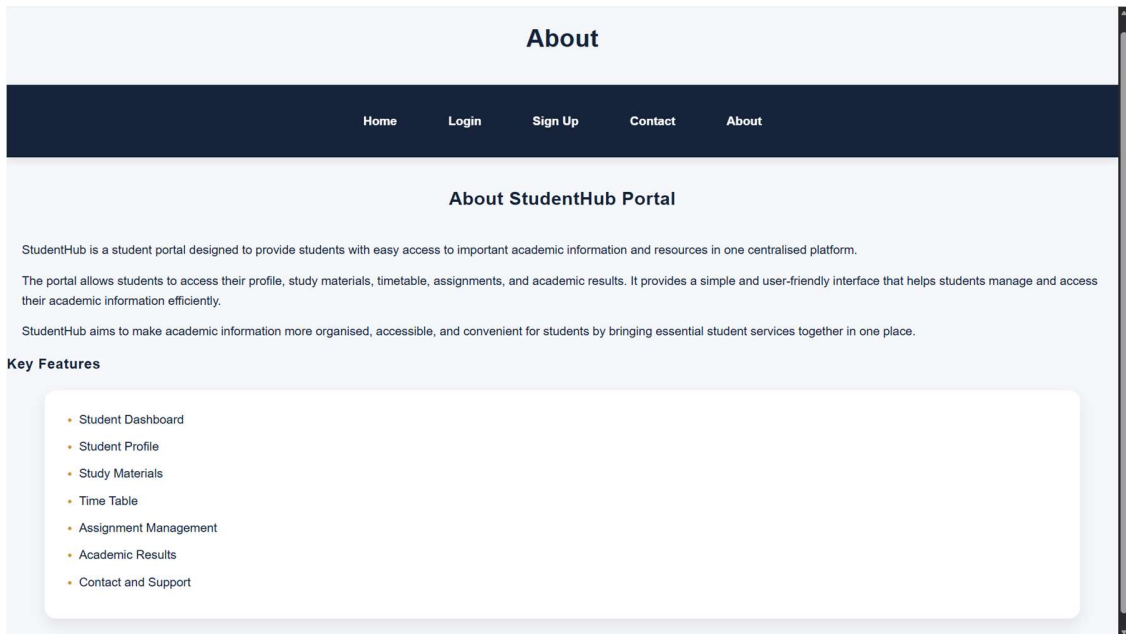
: 9876543210

Department

: Computer Engineering

Edit Profile

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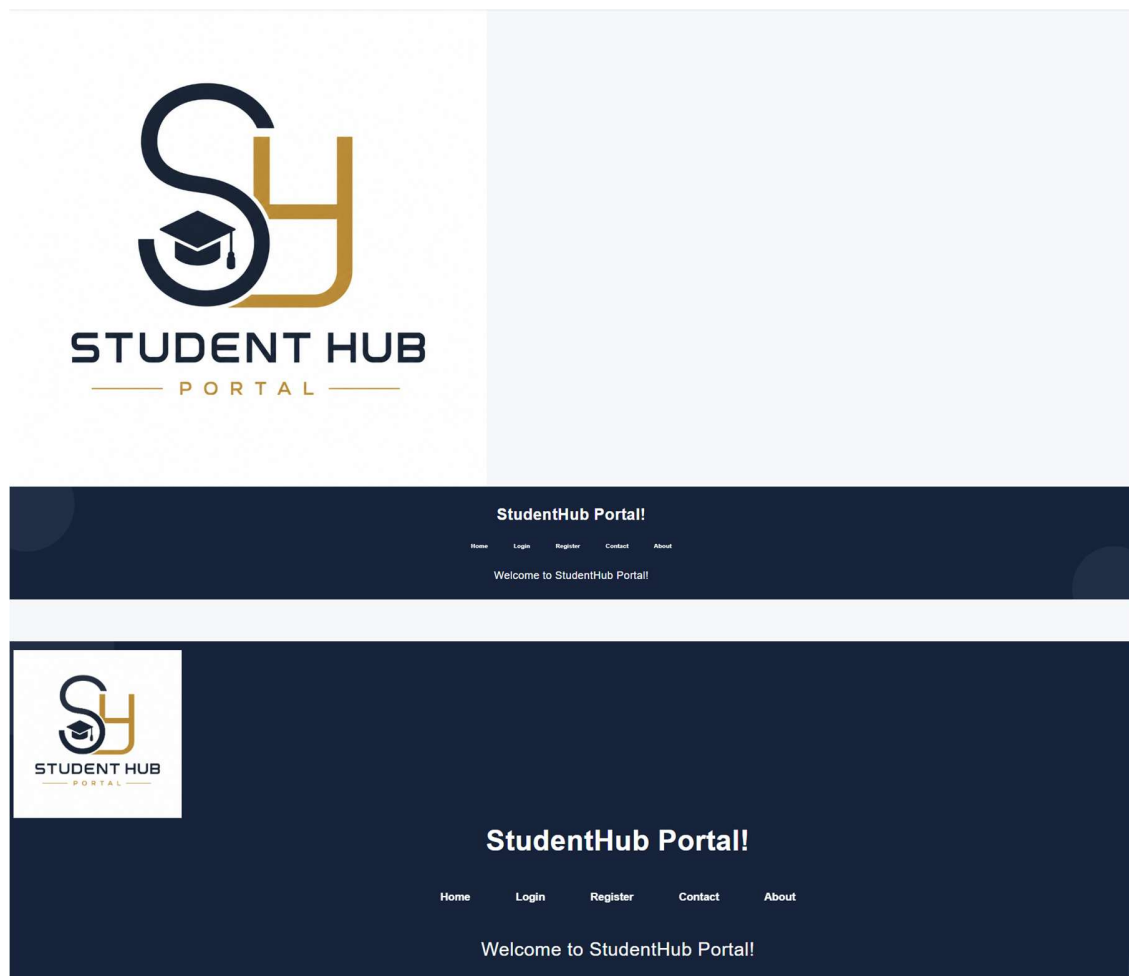


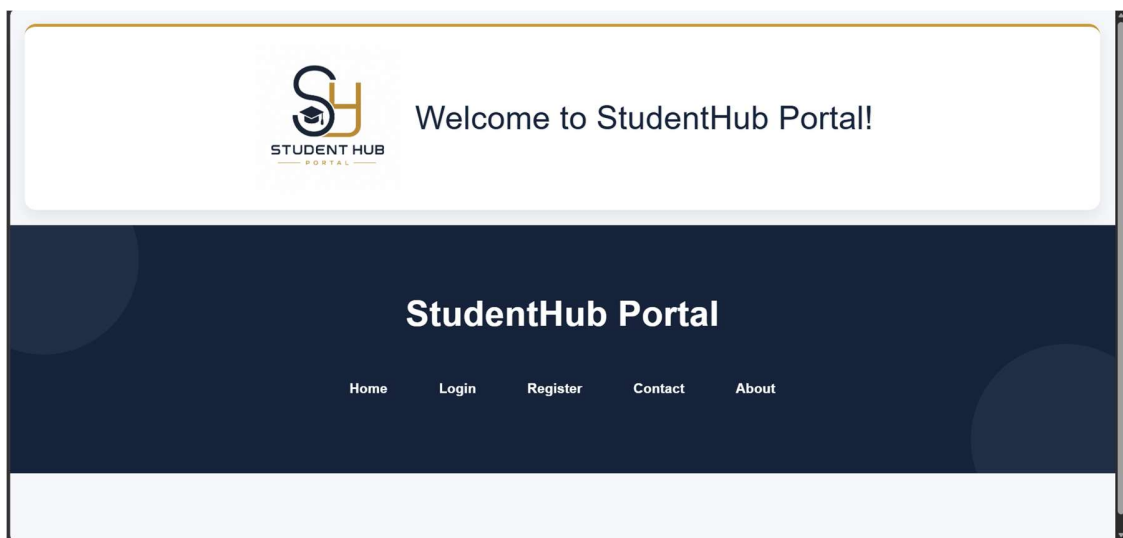
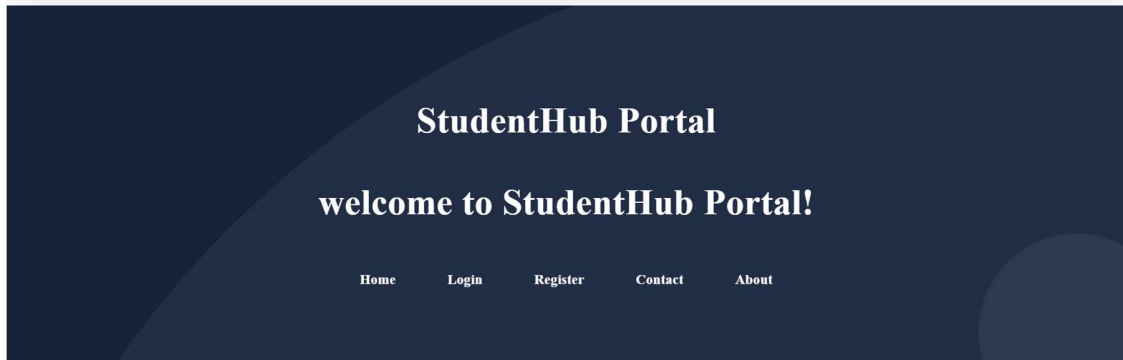
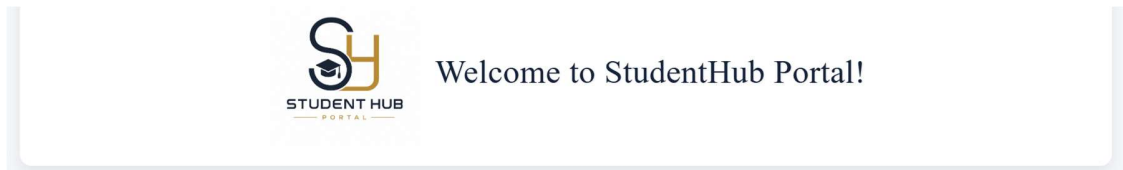
E. GitHub Repository Link

<https://github.com/hkxii/StudentHub-Portal>

F. Challenges Faced

I had faced challenges like making the layout responsive, adjusting spacing for different devices, setting up logo, choosing color combination, etc.





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G. Learning Outcomes

I learned responsive web design, Bootstrap/Tailwind, improved UI design skills, How to solve problems etc.

Faculty Remarks